User Manual

Version 1.0

APpar

Action Potential Parameters Analyzer

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1. Introduction and Scientific Rationale

Action potentials (APs) are the fundamental electrical signals by which neurons encode excitability, transmit information, and reflect the functional state of voltage-gated ion channels. The shape of an action potential is determined by the coordinated activity of sodium, potassium, calcium, and other conductances, together with passive membrane properties.

Because each phase of the AP waveform is governed by distinct physiological mechanisms, quantitative AP analysis is central to studies of neuronal excitability, ion-channel function, disease mechanisms, pharmacological modulation, and firing-pattern adaptation.

Despite the importance of AP waveform analysis, many laboratories continue to rely on partially manual workflows involving cursor placement, spreadsheet calculations, and repeated post-processing steps. Such approaches are time-consuming and may introduce operator-dependent variability.

APpar (Action Potential Parameters Analyzer) was developed to provide a transparent, reproducible, and customizable framework for action potential analysis within the OriginLab Origin/OriginPro environment.

APpar automatically detects action potentials, validates thresholds, calculates electrophysiological parameters, and writes structured results into Origin workbooks while preserving compatibility with established Origin workflows.

The need for reproducible AP analysis is especially clear for threshold determination. AP threshold is not a directly observed physical boundary. It is an operational definition of the membrane voltage at which regenerative depolarization begins according to a chosen criterion. Across the literature, threshold may be defined by a fixed derivative criterion, a relative derivative change, an inflection point, a second derivative criterion, or visual inspection. These choices are not interchangeable. A different threshold definition can change threshold voltage, AP amplitude, rise time, AP duration, AP area above threshold, and other derived measurements.

APpar addresses this issue by making threshold criteria explicit and by implementing backward-search TRUE-threshold validation. The software first detects an initial forward dV/dt threshold crossing, identifies the action potential overshoot, searches backward from the upstroke to the closest preceding dV/dt maximum, and then searches backward to the user-defined derivative threshold crossing. The validated threshold point is used for final calculations, while the forward and backward threshold positions remain available for quality-control review.

APpar also formalizes AP amplitude as overshoot minus threshold voltage. This convention measures the voltage excursion after AP initiation. It is not the same as RMP-to-peak amplitude, peak voltage, or overshoot-minus-undershoot amplitude. Because different laboratories may prefer different conventions, APpar reports the underlying voltage components RMP, threshold, overshoot, and undershoot so that users can calculate alternate amplitude definitions during downstream analysis if needed.

2. Scope of APpar

APpar is intended for quantitative analysis of membrane-voltage recordings containing action potentials. It is useful for current-clamp recordings from neurons, including evoked single AP or repetitive AP firing and spontaneous firing. The default logic was optimized around neuronal AP waveforms, but the analysis settings are intentionally adjustable so that users can adapt detection to different cell types, recording conditions, sampling frequencies, noise levels, and temperatures.

Typical applications include:

• Current-clamp recordings from neurons

• Evoked single AP or repetitive firing

• Spontaneous firing

• Excitability studies

• Ion-channel investigations

• Pharmacological studies

• Disease-model characterization

The default software settings were optimized for recordings from dorsal root ganglion (DRG) neurons at T=23C but can be adapted to other neuronal and excitable cell types through adjustable analysis settings.

APpar is not intended to replace electrophysiological judgment. Automated detection should always be accompanied by visual review of recordings and verification that chosen settings are physiologically appropriate.

3. Software Architecture

APpar exists in two complementary implementations:

LabTalk Version
Native OriginLab LabTalk implementation.

Origin-Python Version
Python implementation launched through a LabTalk wrapper.
The Python implementation reproduces the analytical logic of the LabTalk version while providing improved execution speed for large datasets and long recordings.
The Python implementation uses:
• NumPy
• SciPy
• Pandas
• Origin-Python interface
The active Origin worksheet serves as both the input source and output destination.
Input:

Column A = Time
Columns B–N = Voltage traces
Output:
• Raw dV/dt columns
• Smoothed dV/dt columns
• Results workbook containing AP measurements

4. Installation

Software Requirements
Windows 10 or Windows 11
Origin 2022b or later
OriginPro 2025 supported
Required Python packages:
• numpy
• scipy
• pandas

Python Package Installation
Open from Origin menu:
Connectivity → Python Packages
Install:
numpy
scipy
pandas
Then open:
Connectivity → Python Console
Execute:
```
import sys
print(sys.executable)
import subprocess
subprocess.check_call([sys.executable, "-m", "pip", "install", "numpy", "scipy",
"pandas"])
```
Restart Origin.

File Placement
Place:
APpar.ogs
APpar.py
into:
C:\Users**USERNAME**\Documents\OriginLab\User Files
Edit APpar.ogs and replace:
C:\Users**dv93**\Documents\OriginLab\User Files\APpar.py
with the path corresponding to your Windows username.
Restart Origin.

5. Input Data Requirements

The active Origin worksheet must contain:
Column A:
Time values
Columns B–N:
Membrane-voltage traces
Important:
ALL TIME VALUES MUST BE IN SECONDS
ALL VOLTAGE VALUES MUST BE IN VOLTS
Failure to use seconds and volts will result in incorrect calculations.
Recommended structure:

Column	Content
A	Time (s)
B	Voltage Trace 1 (V)
C	Voltage Trace 2 (V)
D	Voltage Trace 3 (V)

and so forth.

6. Running APpar

1. Activate the worksheet containing AP traces.
2. Open Origin Command Window: Window → Command Window or Alt+3
3. Type: APpar
4. Press Enter.
5. The settings dialog appears.
6. Enter analysis parameters.
7. Click OK.
8. APpar performs analysis.
9. Results workbook is created automatically.

7. User-Defined Settings

Number_of_Traces
Number of voltage traces to analyze.

RMP_Start_Time
Beginning of resting membrane potential window.
Default: 0.004 s

RMP_End_Time
End of resting membrane potential window.
Default: 0.009 s

AP_Start_Time
Beginning of AP search region.

Default: 0.010 s

AP_End_Time
End of AP search region.
Default: End of recording.

dVdt_Threshold_V_per_s
Derivative threshold used for AP detection.
Default: 8 V/s

dVdt_Filter_Number_of_Points
Derivative smoothing window.
Default: 6 points

Minimal_AP_Overshoot
Minimum acceptable AP peak.
Default: 0 V

Minimal_AP_Duration
Minimum AP duration.
Default: 0.001 s

Minimal_Inter_AP_Interval
Minimum interval between adjacent APs.
Default: 0.010 s

Minimum_dVdt_Threshold_Duration
Minimum duration dV/dt must remain above threshold.
Default: 0.0002 s

Data_Writing_Pause
Pause inserted during workbook writing.
Default: 0.025 s

8. Core Detection Algorithm

APpar performs the following steps:

1. Read voltage traces.
2. Calculate raw dV/dt using central-difference approximation.
3. Smooth dV/dt.
4. Search for derivative threshold crossings.
5. Apply interspike interval filter.
6. Search for overshoot.
7. Verify local peak.
8. Apply duration criteria.
9. Apply overshoot criteria.

10. Perform TRUE-threshold validation.
11. Calculate AP parameters.
12. Write results.

The central-difference derivative is: dV/dt(i) = [V(i+1)-V(i-1)] / [t(i+1)-t(i-1)]. This method reduces directional bias and provides a robust estimate of AP upstroke velocity.

9. TRUE-Threshold Validation Algorithm

The TRUE-threshold algorithm is the defining feature of APpar.
Traditional workflows frequently accept the first forward derivative threshold crossing as AP threshold.
APpar performs additional validation.
Workflow:

1. Detect forward dV/dt threshold crossing.
2. Locate AP overshoot.
3. Locate nearest preceding dV/dtMAX.
4. Search backward until dV/dt falls below user criterion.
5. Select next point as TRUE threshold.
6. Recalculate all AP parameters.

The results table reports:
• Forward threshold row
• Forward threshold time
• TRUE threshold row
• TRUE threshold time
• Threshold source
• Forward vs TRUE row difference
This allows complete auditability of threshold determination.

10. Parameter Definitions and Interpretation

RMP. Resting membrane potential measured from user-defined baseline window.

Threshold Voltage. Validated AP initiation voltage.

dV/dt at Threshold. Derivative value at threshold.

Overshoot. Maximum membrane voltage during AP.

Undershoot. Most negative voltage after AP peak.

AP Amplitude. Overshoot − Threshold Voltage

Half Amplitude. 50% of AP amplitude.

Rise Time. Threshold → Overshoot interval.

Decay Time. Overshoot → Return-to-threshold interval.

AP Duration. Threshold-to-threshold duration.

Half Width. Duration measured at half amplitude.

Width at 0 mV. Time spent above 0 mV.

Area Above Threshold. $\int(V_m - V_{threshold})dt$

dV/dtMAX. Maximum derivative.

dV/dtMIN. Minimum derivative.

ISI. Threshold-to-threshold interval between consecutive APs.

11. Results Workbook Structure

The results workbook contains:

Analysis Settings. Stored at top of worksheet.

AP-by-AP Measurements. One row per action potential.

Trace Summary. Number of APs detected.

Threshold Audit Information. Forward threshold values; TRUE threshold values; Threshold source; Threshold row difference

12. Recommended Workflow

1. Import traces into Origin.
2. Verify units.
3. Run APpar.
4. Inspect dV/dt traces.
5. Review threshold positions.
6. Review AP count.
7. Export results.
8. Perform statistical analysis.
9. Archive settings with data.

13. Troubleshooting

No APs Detected

Possible causes:

• dV/dt threshold too high
• incorrect units

• AP search window incorrect
• excessive smoothing

Too Many APs Detected
Possible causes:
• dV/dt threshold too low
• noise
• insufficient inter-AP interval

Incorrect AP Parameters
Verify:
• seconds are used for time
• volts are used for voltage
• baseline window is appropriate
• AP search window covers APs

Program Does Not Start
Verify:
• APpar.py exists
• APpar.ogs path is correct
• Python packages are installed
• Origin was restarted

14. Figure-by-Figure Guide

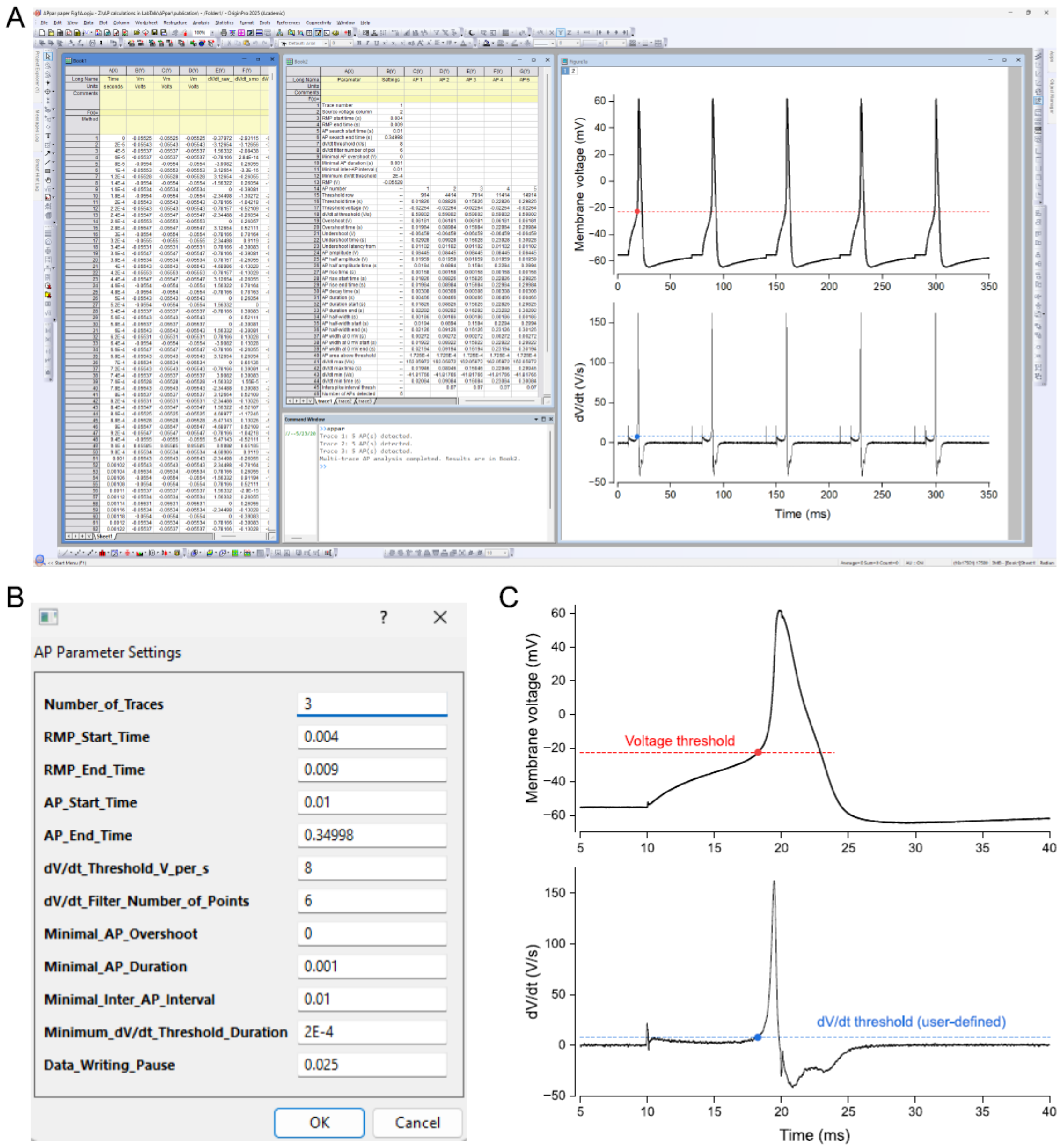

Figure 1. APpar software overview. Representative Origin/OriginPro workflow for APpar. The active worksheet contains time and voltage recordings (A), while the settings dialog window (B) defines analysis parameters including number of traces, RMP interval, AP search interval, dV/dt threshold, derivative smoothing, overshoot filter, AP duration filter, inter-AP interval, and minimum dV/dt threshold duration. Output includes derivative traces and structured AP parameter tables (A, C).

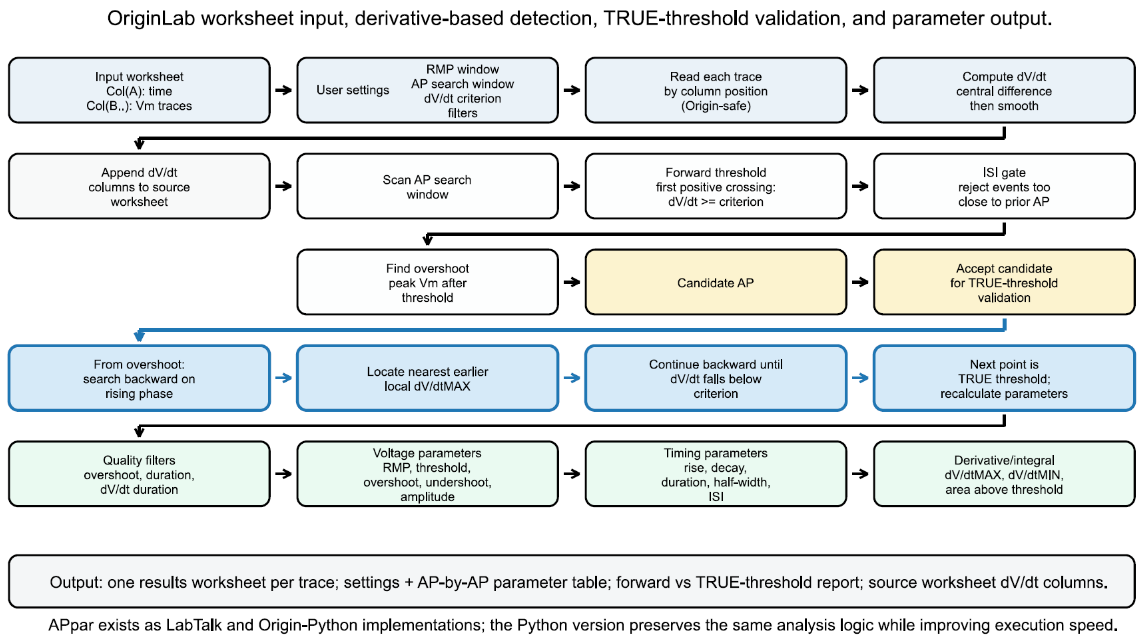

Figure 2. APpar operational workflow and AP detection algorithm. Operational diagram of APpar based on the LabTalk and Origin-Python code structure. The workflow includes worksheet import, dV/dt calculation and smoothing, forward threshold detection, inter-AP interval filtering, overshoot detection, TRUE-threshold validation, AP parameter extraction, quality filtering, and output generation. Blue arrows indicate iterative threshold-validation steps.

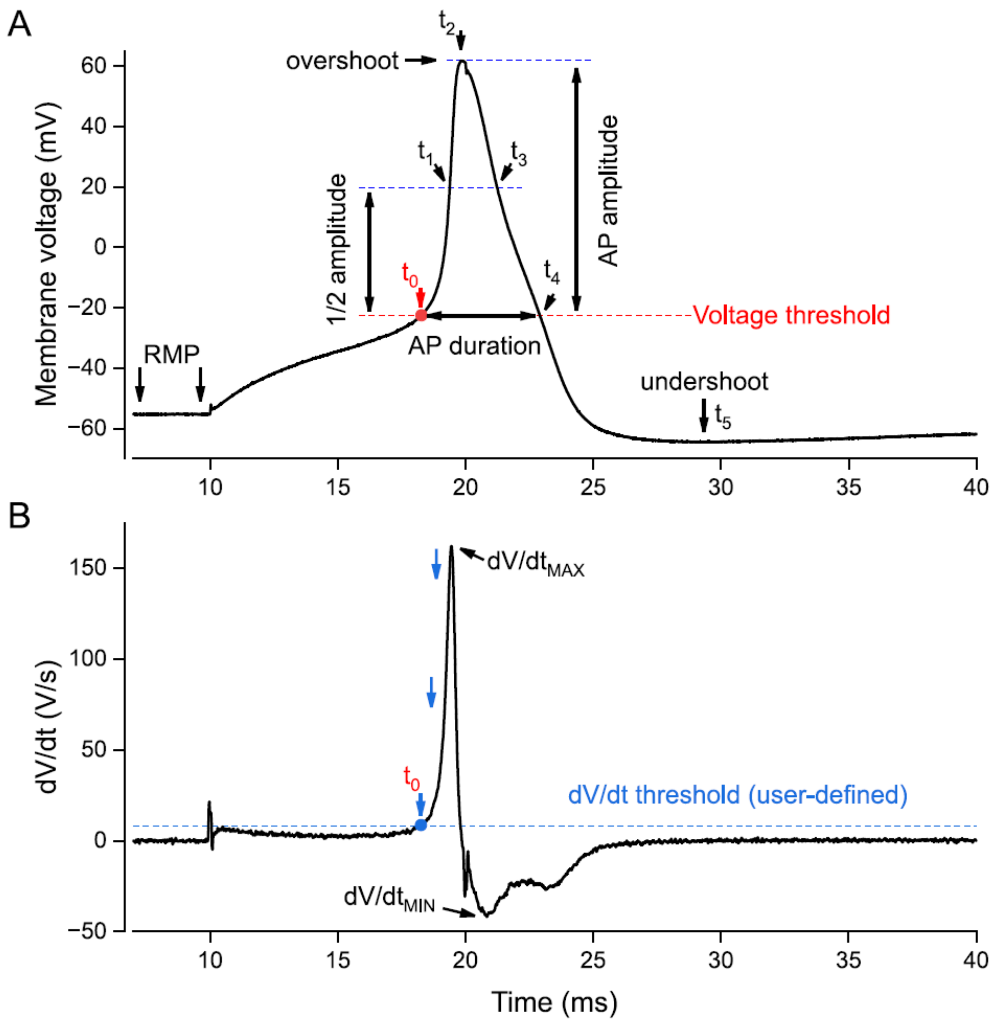

Figure 3. Definition of AP parameters and TRUE-threshold validation. Representative AP waveform (A) and derivative (B) trace showing the parameters calculated by APpar, including threshold, overshoot, undershoot, AP amplitude, AP duration, AP half-width, dV/dt_{MAX}, and dV/dt_{MIN}. Time points t_0-t_5 define key stages of the AP waveform. t_0 indicates validated AP threshold, t_1 rising-phase half-amplitude crossing, t_2 AP overshoot, t_3 falling-phase half-amplitude crossing, t_4 return to threshold voltage, and t_5 AP undershoot.

The lower panel (B) illustrates TRUE-threshold validation. After initial forward dV/dt threshold detection, APpar searches backward from AP overshoot to the nearest earlier local dV/dt maximum and then to the user-defined derivative threshold crossing. Blue arrows indicate the evolution of the algorithmic search process.

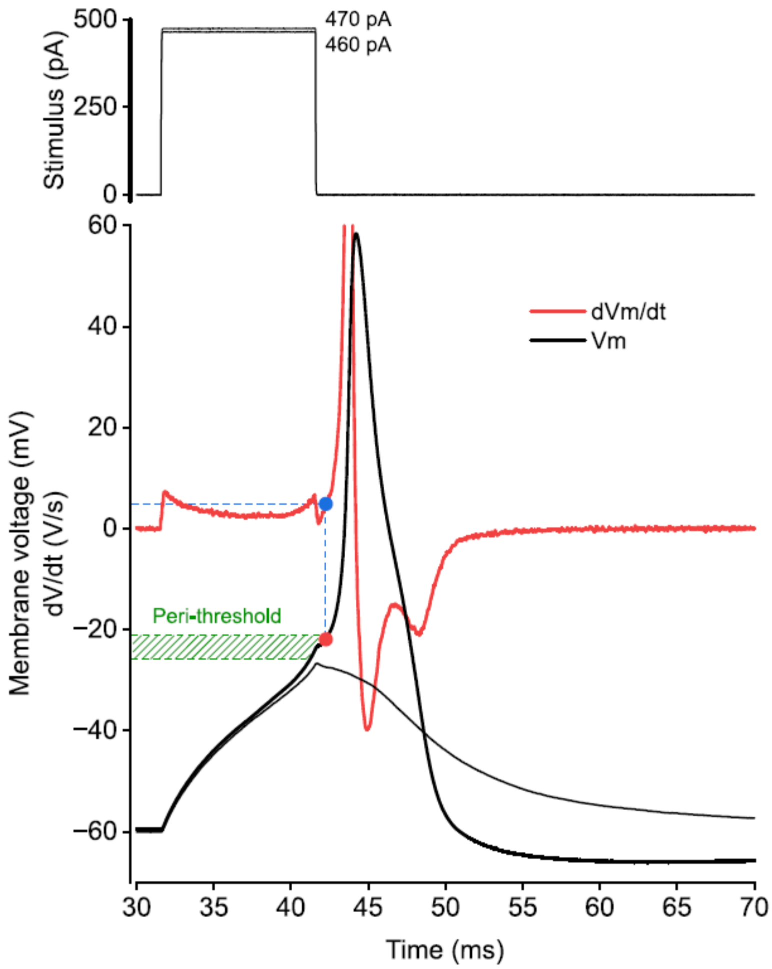

Figure 4. Rheobase, peri-threshold region, and AP threshold voltage. Representative recordings illustrating rheobase determination and peri-threshold membrane behavior in a small DRG neuron. Current injection of 460 pA (upper panel) did not evoke an action potential, whereas 470 pA evoked an AP (lower panel), defining rheobase under these recording conditions. The lower panel shows the evoked AP overlayed with its derivative and highlights the peri-threshold region (green) associated with AP initiation and threshold voltage determination.

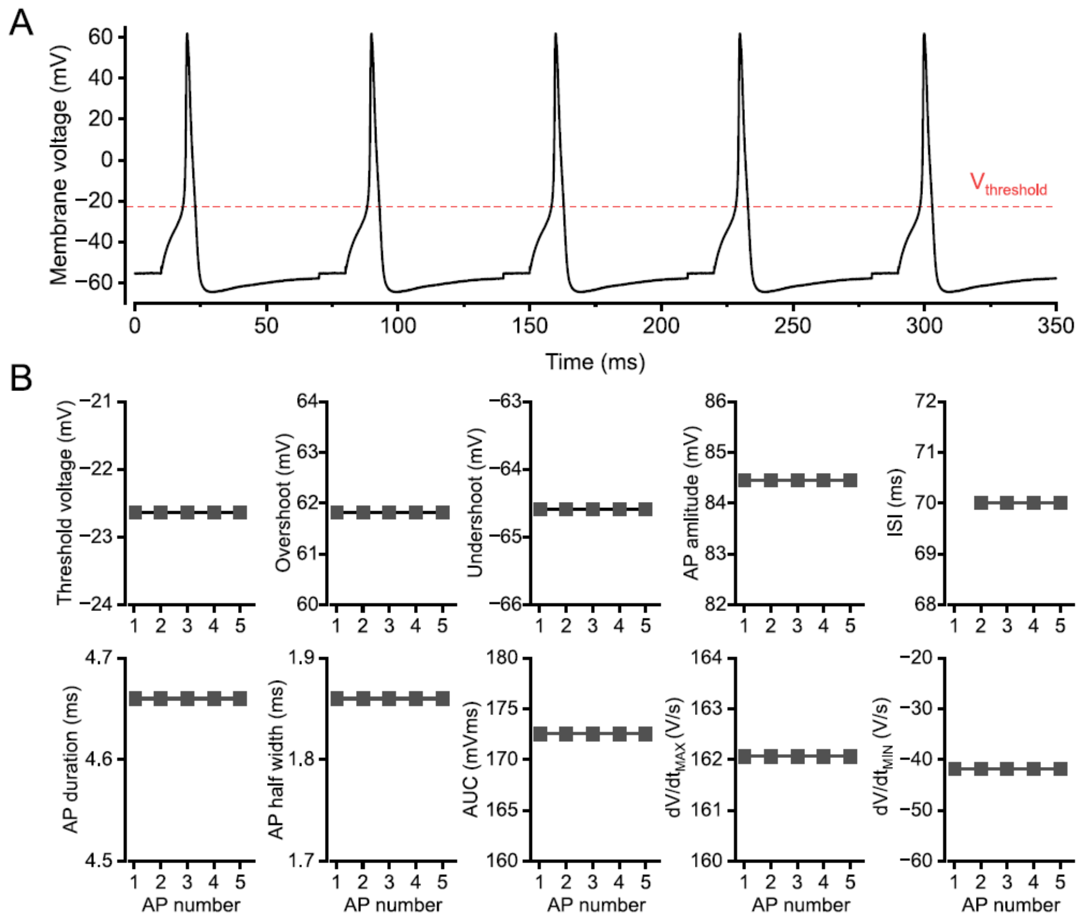

Figure 5. Stability of calculations performed on identical copied APs. Repeated analysis of identical copied AP waveforms (A) demonstrates stability of APpar parameter extraction (B). Threshold voltage, overshoot, undershoot, AP amplitude, ISI, AP duration, AP half-width, AP area above threshold, dV/dt_{MAX}, and dV/dt_{MIN} remain stable across AP number when the input waveform is identical.

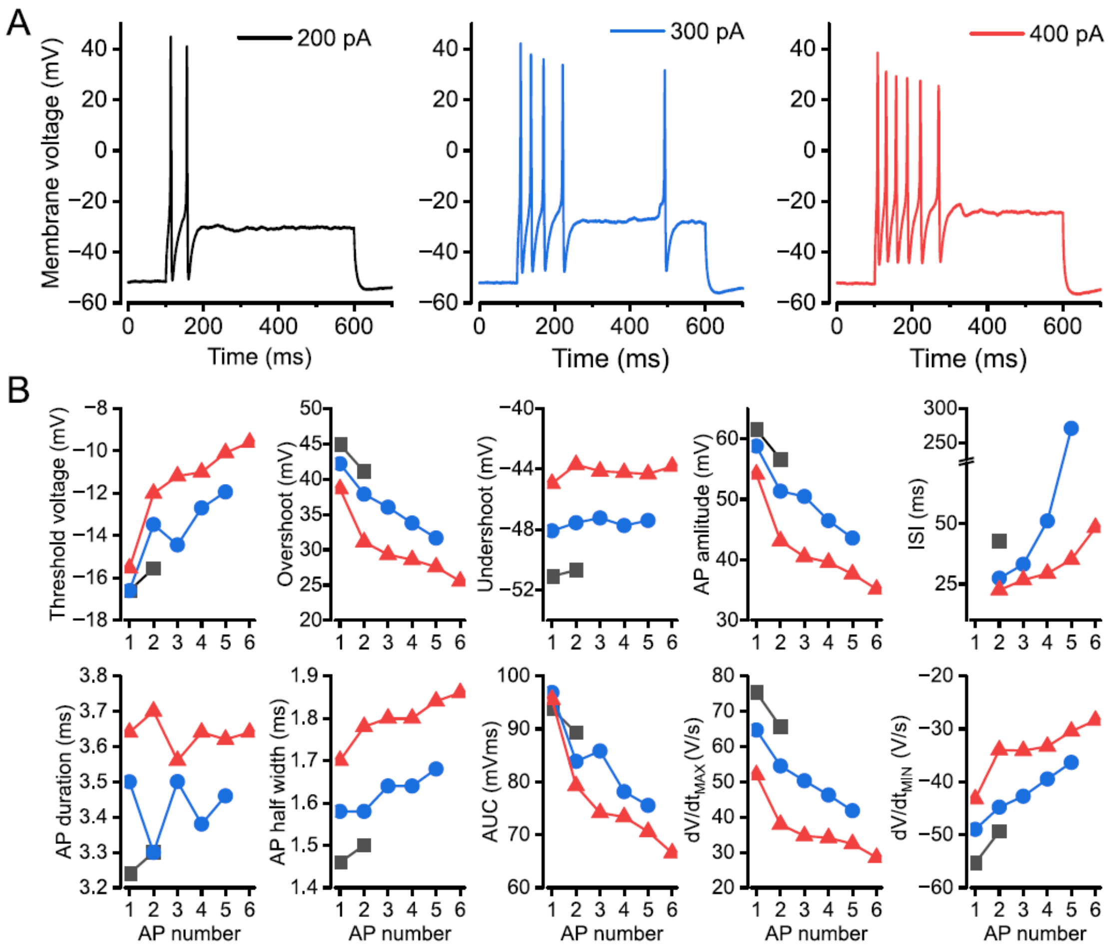

Figure 6. Representative recordings from a small DRG neuron. Representative current-evoked AP trains recorded from a small DRG neuron at different depolarizing current injections, 200 pA, 300 pA and 400 pA denote respective stimulus strength (A). APpar extracts AP-by-AP parameter trajectories including threshold voltage, overshoot, undershoot, AP amplitude, ISI, AP duration, AP half-width, AP area above threshold, dV/dt$_{MAX}$, and dV/dt$_{MIN}$ (B).

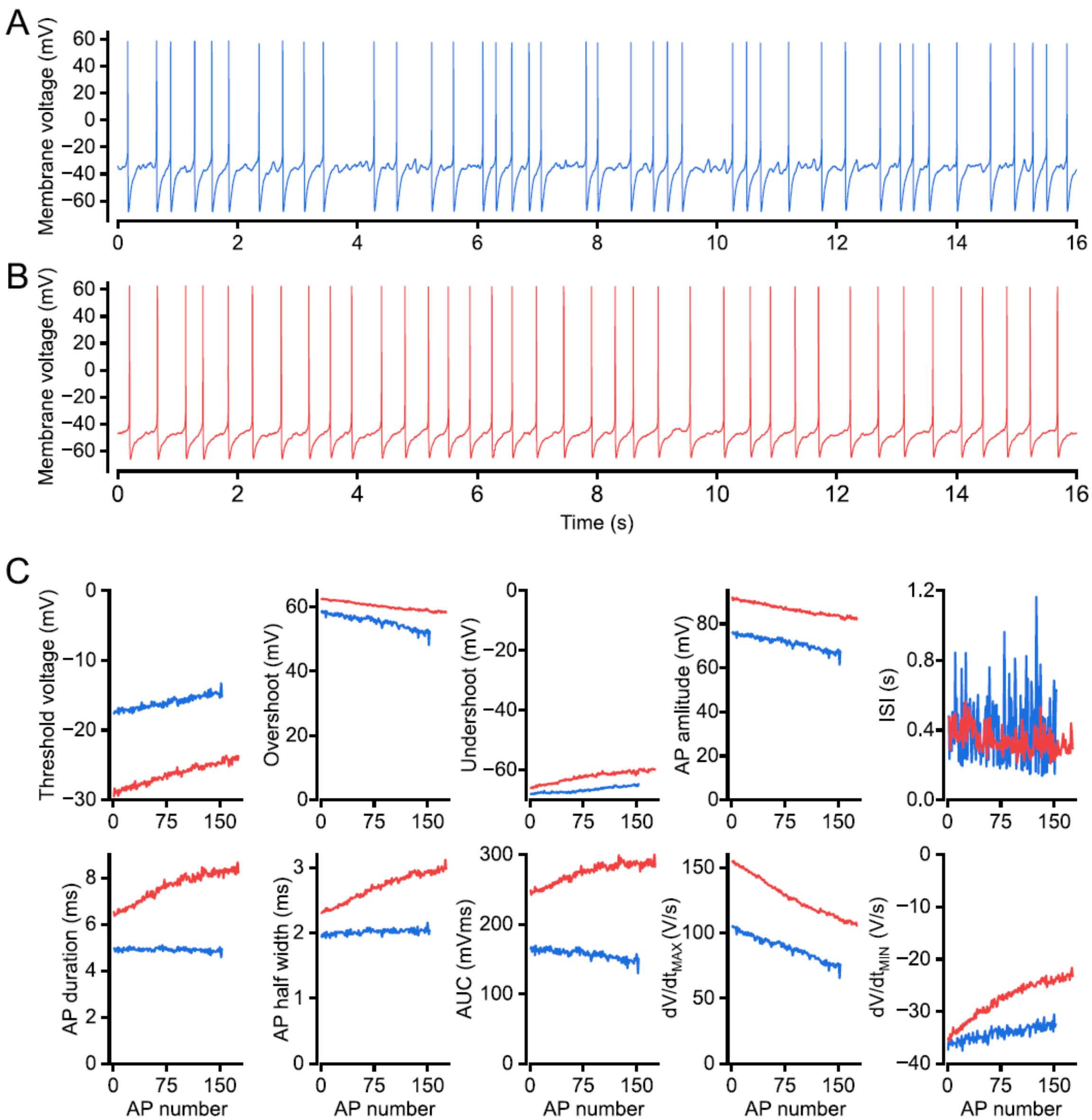

Figure 7. Representative recordings of spontaneous APs in two small DRG neurons. Long-duration spontaneous AP recordings from two small DRG neurons (A, B). The first 16 s are shown in the top panels, while AP parameters were calculated from 60 s recordings. APpar quantifies dynamic changes in threshold voltage, overshoot, undershoot, AP amplitude, ISI, AP duration, half-width, AP area above threshold, dV/dt_{MAX}, and dV/dt_{MIN} across spontaneous firing sequences (B). The color-code is the same across all corresponding panels and graphs.

15. References

Vasylyev DV and Waxman SG (2026) APpar: automated action potential parameter analysis software for reproducible electrophysiological measurements in neurons. bioRxiv